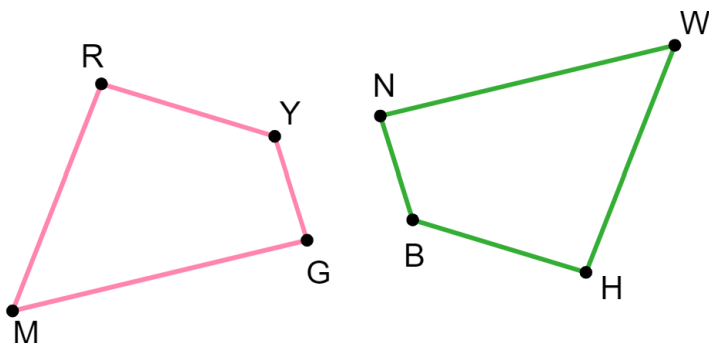




5.9.5 Wrap-Up: Error Analysis

1. Quadrilaterals $RYGM$ and $HBNW$ are shown.



$RYGM \cong HBNW$ and the following information is also true.

$$m\angle RYG = (10x - 6)^\circ, MG = 2k + 3, NW = 4k - 29, \\ m\angle MGY = (7x - 5)^\circ, m\angle BNW = 86^\circ, HB = k + 9$$

Hector used the given information to write the two equations shown.

$$10x - 6 = 86$$

$$4k - 29 = 2k + 3$$

One of Hector's equations is NOT correct. Determine which one, and correct his error.

Unit 5, Lesson 10: Congruence in Two-Dimensional Figures – Part 2



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve real-world problems involving congruence.



5.10.1 Warm-Up: Notice and Wonder

A table of menu prices is shown.

Burgers	Chips	Fries
Deluxe Burger	\$6.90	\$7.70
Deluxe Cheese	\$7.70	\$8.45
Bacon Burger	\$7.95	\$8.65
Bacon Cheese	\$8.25	\$8.95
Double Burger	\$8.35	\$9.30
Double Cheese	\$9.25	\$9.95
Double Bacon Cheese	\$9.50	\$10.25
Cheese Garden Burger	\$8.25	\$8.85
Mushroom Swiss	\$8.25	\$8.95
Jalapeno Pepper Jack	\$8.25	\$8.95
BBQ Western Bacon	\$9.20	\$9.85
Pastrami Burger	\$9.20	\$9.85
Power Pac	\$10.45	\$11.05
Cheese Power Pac	\$11.25	\$11.70

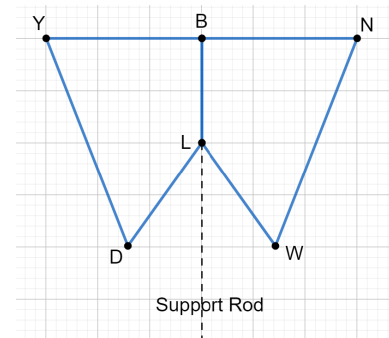
- What do you notice about prices of the burgers with cheese?
- What do you notice about ordering chips versus fries?
- There is a price difference between adding chips or fries with a double burger and adding chips or fries with cheese garden burger. Do you think this means that when you order a double burger you get more fries?
- When you look at a menu, do you consider the price differences or just order what you would like to eat?



5.10.2 Guided Instruction: Corresponding Parts in the Real World

Oscar and Aarna worked together to create a prototype obstacle for a drone obstacle course. They used grid paper to create their design as shown.

To be sure the design met regulations for the competition, they used the equations to input the design into a computer program. They discovered that some of their work disappeared, so they used the given equations knowing that $YBLD \cong NBLW$.



The perimeter of $YBLD$ is 293 centimeters (cm). $\angle YBL$ is a right angle.

$$YB = 75 \text{ cm} \quad WL = (3x + 10) \text{ cm} \quad m\angle LWN = (4k + 9)^\circ$$

$$m\angle DLB = (12k + 1)^\circ \quad WN = (6x + 5) \text{ cm} \quad LB = 50 \text{ cm}$$

$$m\angle BYD = (5k + 8)^\circ$$

- What is the value of x ?

2. What is the length of WL ? Include units.

3. What is the length of YD ? Include units.

4. What is the value of k ?

5. What is the measure of $m\angle BYD$, in degrees?

6. What is the measure of $m\angle LDY$, in degrees?

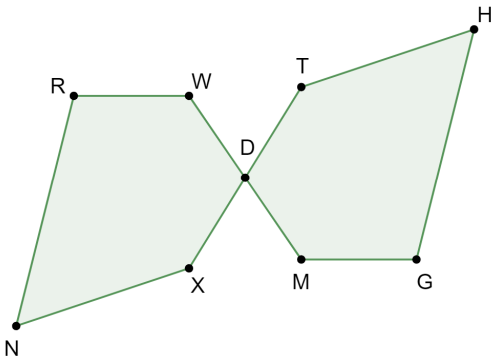


5.10.3 Collaboration: Corresponding Parts in the Real World

Work with your partner to complete the following.

French formal gardens use geometric shapes that repeat throughout the garden. Dajon is a landscape designer. He is designing a garden in the tradition of a French

formal garden. While at the client's home, he wrote down some measurements, in feet (ft), and then sketched polygons $RWDXN$ and $DTHGM$ to represent the garden. In the tradition of the congruence of polygons found in French formal gardens, Dajon also wrote $NRWDX \cong HGMDT$.



The garden's measurements are given in the table.

Garden Measurements	
$WD = (4x - 0.1) \text{ ft}$	$MD = 10.3 \text{ ft}$
$RW = 12 \text{ ft}$	$TH = (8x - 1.8) \text{ ft}$
$DX = (3x + 3.4) \text{ ft}$	$HG = (10x - 1.3) \text{ ft}$

1. Determine the perimeter of the garden. Show your work.



Once the garden has been installed, Dajon will check the angles of the polygons.

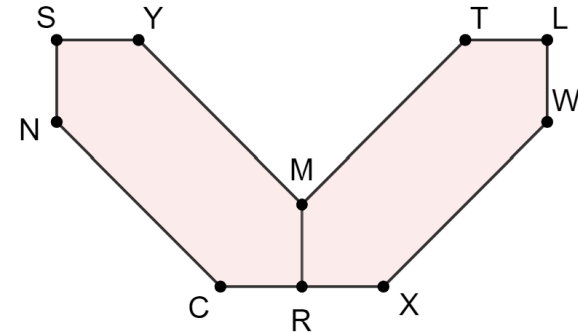
$m\angle HTD = (5k - 6)^\circ$
$m\angle XNR = 58^\circ$
$m\angle NRW = (4k + 4)^\circ$
$m\angle DMG = (6k - 20)^\circ$
$m\angle MGH = 104^\circ$

- Determine what measurement, in degrees, Dajon should use to check $m\angle XDW$. Show your work.



5.10.4 Your Turn!

- Mireia is learning how to make stained glass in art class. She has created a design and labeled the vertices to help her keep track of the measurements. Her design is shown.



Mireia's design was created so that $RMYSNC \cong RMTLWX$, $\overline{CN} \cong \overline{MY}$, and $\overline{CR} \cong \overline{RM} \cong \overline{YS} \cong \overline{SN}$. Mireia needs to determine how much solder she needs. Solder is used along the perimeter, in inches (in.), of the design to set the glass in place.

Use the given measurements, in inches, to determine the amount of solder.

$$WX = 9x - 2.3$$

$$YM = 6x + 1.3$$

$$MR = 3x - 0.6$$



5.10.5 Wrap-Up: Reflection

1. Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentence by checking all that apply.

I don't understand

- ☐ SSS
- ☐ SAS
- ☐ ASA
- ☐ AAS
- ☐ HL
- ☐ CPCTC

... YET.

- ☐ Naming congruent parts
- ☐ Paragraph proofs
- ☐ Two-column proofs
- ☐ Flowchart proofs
- ☐ Finding values in congruent polygons

do(does) NOT make sense ... YET.

I can explain

- ☐ SSS.
- ☐ SAS.
- ☐ ASA.
- ☐ AAS.
- ☐ HL.
- ☐ CPCTC.
- ☐ naming congruent parts.
- ☐ finding values in congruent polygons.

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